

1. Consider **(P)** given below:

$$\begin{aligned} & \text{Minimize} && -x_1 + 2x_2 - x_3 \\ & \text{subject to} && -2x_1 + x_2 - x_3 \geq 1 \\ & && x_1 - 2x_2 + 3x_3 \leq 2 \\ & && x_1 \geq 0, x_2 \geq 0, x_3 \geq 0. \end{aligned}$$

[2pnts.]

(a) Write the dual of **(P)**.

[3pnts.]

(b) Using duality theory check if **(P)** has an optimal solution (you do **not** have to give the optimal solution).

Sol: (a) The Dual is

$$\begin{aligned} & \text{Minimize} && -y_1 + 2y_2 \\ & \text{subject to} && 2y_1 + y_2 \geq 1 \\ & && -y_1 - 2y_2 \geq -2 \\ & && 3y_1 + y_2 \geq 1 \\ & && y_1 \geq 0, y_2 \geq 0. \end{aligned}$$

or

$$\begin{aligned} & \text{Maximize} && y_1 - 2y_2 \\ & \text{subject to} && -2y_1 - y_2 \leq -1 \\ & && y_1 + 2y_2 \leq 2 \\ & && -3y_1 - y_2 \leq -1 \\ & && y_1 \geq 0, y_2 \geq 0. \end{aligned}$$

(b) Sol: $[0, 1, 0]^T \in \text{Fea}(P)$ and $[0, 1]^T \in \text{Fea}(D)$ therefore since both have feasible solutions so both have optimal solutions by duality theory.

There are multiple answers for this question.

2. Consider the LPP **(P)** of the form,

$$\begin{aligned} & \text{Maximize} && \mathbf{c}^T \mathbf{x} \\ & \text{subject to} && A_{4 \times 3} \mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}. \end{aligned}$$

Let $\mathbf{x}_0 = [0, r, s]^T$ ($r, s > 0$), be the unique optimal solution of **(P)** and let $\mathbf{y}_0$ be optimal for the Dual of **(P)**.

[3pnts.]

(a) If $\mathbf{c}$ is changed to $\mathbf{c} + [-1, 0, -1]^T$ then is $\mathbf{y}_0$ optimal for the Dual of new **(P)**? Justify.

[3pnts.]

(b) If the first column $\tilde{\mathbf{a}}_1$ of A is changed to $\tilde{\mathbf{a}}_1 + [1, 2, 0, 1]^T$ then is $\mathbf{y}_0$ optimal for the Dual of new **(P)**? Justify.

[2pnts.]

(c) If $\mathbf{b} = [2, 1, 3, 0]^T$ and the first row $\mathbf{a}_1^T$ of A is changed to $\mathbf{a}_1^T + [0, 2, 0]$ then will new **(P)** have an optimal solution? Justify.

(All the parts in the above question are independent)

Soln: The Dual is of the form

$$\begin{aligned} &\text{Minimize } \mathbf{b}^T \mathbf{y} \\ &\text{subject to } A^T \mathbf{y} \geq \mathbf{c}, \mathbf{y} \geq \mathbf{0}. \end{aligned}$$

(a) **Sol:** Since the feasible region of $(\mathbf{P})$ and $\text{new}(\mathbf{P})$ are same so $\mathbf{x}_0$ is feasible for the new $(\mathbf{P})$ as well.

Note that since $\mathbf{x}_0 = [0, r, s]^T$ ($r, s > 0$), $\mathbf{y}_0$ are optimal for $(\mathbf{P})$, Dual of $(\mathbf{P})$ respectively, and $r, s > 0$, so by the complementary slackness theorem the second and the third constraints of the Dual of $(\mathbf{P})$ must be satisfied as an equality by $\mathbf{y}_0$, that is, $(A^T \mathbf{y}_0)_2 = c_2$ and $(A^T \mathbf{y}_0)_3 = c_3$.

But in the constraints of the new Dual the third constraint of the new Dual is not satisfied as an equality by $\mathbf{y}_0$ (although $\mathbf{y}_0$ is feasible for the new Dual, check this) so atleast one of $\mathbf{x}_0, \mathbf{y}_0$ is **not** optimal for the new $(\mathbf{P})$, the Dual of the new $(\mathbf{P})$, respectively. [1] mark

Case 1: $\mathbf{y}_0$ is not optimal for the Dual of new $(\mathbf{P})$.

Case 2: $\mathbf{x}_0$ is not optimal for new $(\mathbf{P})$ and suppose $\mathbf{x}'$ is the new optimal then it must be a feasible solution of $(\mathbf{P})$. Since $\mathbf{x}_0$ is the **unique** optimal for $(\mathbf{P})$ so $\mathbf{b}^T \mathbf{y}_0 = \mathbf{c}^T \mathbf{x}_0 > \mathbf{c}^T \mathbf{x}' \geq (\mathbf{c}^T + [-1, 0, -1])\mathbf{x}'$ which implies $\mathbf{y}_0$ is not optimal for the Dual of the new $(\mathbf{P})$.

So in either case $\mathbf{y}_0$ is not optimal for the Dual of the new $(\mathbf{P})$ [2] marks

No marks awarded for concluding **wrongly** that $\mathbf{y}_0$ is optimal for the Dual of the new $(\mathbf{P})$.

(b) **Sol:**

Part 1: Check that (by showing all the required calculations and arguments) that $\mathbf{x}_0, \mathbf{y}_0$ is feasible for the new $(\mathbf{P})$ and the Dual of the new $(\mathbf{P})$, respectively.

Part 2: Also since $\mathbf{x}_0, \mathbf{y}_0$ are optimal for $(\mathbf{P})$, Dual of $(\mathbf{P})$ respectively, by the fundamental theorem of duality $\mathbf{b}^T \mathbf{y}_0 = \mathbf{c}^T \mathbf{x}_0$. Since $\mathbf{b}, \mathbf{c}$ has not changed for the new problem so from Part 1 and the above argument $\mathbf{x}_0, \mathbf{y}_0$ are optimal for $(\mathbf{P})$ and the Dual of new $(\mathbf{P})$, respectively.

Only [1] mark given if any of the above parts are missing or the required calculations are not shown but the conclusion is correct.

No marks awarded if the conclusion is wrong.

(c) **Sol:** Note that $\mathbf{0} \in \text{Fea}(\text{new } P)$ (you have to show this) and $\mathbf{y}_0$ is feasible for the Dual of the new $(\mathbf{P})$ (you have to show this) so new $(\mathbf{P})$ (and dual of new $(\mathbf{D})$) has optimal solution by duality theory.

No part marks for this part.

3. Consider the following LPP $(\mathbf{P})$:

$$\begin{aligned} &\text{Minimize } \mathbf{c}^T \mathbf{x} \\ &\text{subject to } \mathbf{A}_{2 \times 4} \mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}, \end{aligned}$$

where $\tilde{\mathbf{a}}_1 = [2, 1]^T, \tilde{\mathbf{a}}_2 = [1, 3]^T, \tilde{\mathbf{a}}_3 = [-5, 0]^T, \tilde{\mathbf{a}}_4 = [-3, 1]^T$, are the columns of A , $\mathbf{b} = [-5, 5]^T, \mathbf{c} = [-1, -4, 6, 2]^T$.

[4pnts.]

(a) Give all the entries of the simplex table (including the $c_j - z_j$ values) for the **BFS** $\mathbf{x}_0$ which has x_1, x_4 as basic variables.

[4pnts.]

(b) Using the **simplex algorithm** either obtain the optimal table and the optimal solution of $(\mathbf{P})$ or conclude that $(\mathbf{P})$ does not have an optimal solution.

Show the necessary calculations for part(a). Give the entering and the leaving variable and pivot element wherever applicable.

Sol: (a)

$c_j - z_j$	0	-4	3	0	
	$B^{-1} \tilde{\mathbf{a}}_1$	$B^{-1} \tilde{\mathbf{a}}_2$	$B^{-1} \tilde{\mathbf{a}}_3$	$B^{-1} \tilde{\mathbf{a}}_4$	$B^{-1} \mathbf{b}$
x_1	1	2	-1	0	2
x_4	0	1	1	1	3

All the required calculations has to be shown.

Sol: (b) In the previous table x_1 is the entering and x_2 is the leaving variable, the pivot element is 2. The optimal table is given by:

$c_j - z_j$	2	0	1	0	
	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\tilde{\mathbf{a}}_3$	$B^{-1}\tilde{\mathbf{a}}_4$	$B^{-1}\mathbf{b}$
x_2	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	1
x_4	$-\frac{1}{2}$	0	$\frac{3}{2}$	1	2

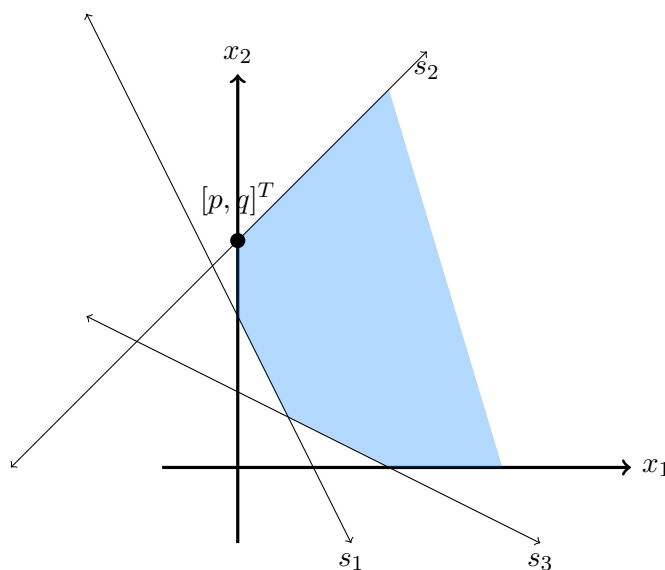
An optimal solution of $(\mathbf{P})$ is $x_1 = 0, x_2 = 1, x_3 = 0, x_4 = 2$.

4. Consider LPP $(\mathbf{P})$ of the form,

$$\text{Minimize } \mathbf{c}^T \mathbf{x}$$

$$\text{subject to } A_{3 \times 2} \mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0},$$

with the shaded feasible region given below. Non negative variable $s_i, i = 1, 2, 3$ is added to the i th constraint of $A_{3 \times 2} \mathbf{x} \leq \mathbf{b}$ and the hyperplanes corresponding to each of the added variables s_i , are marked accordingly in the picture given below.



- [2pnts.] (a) Give the **basic** and the **non basic** variables of the **BFS** corresponding to $[p, q]^T$.
- [3pnts.] (b) Give all **degenerate BFS** in $(\mathbf{P})$ (after adding variables s_i) **if** there are any. Justify.
- [4pnts.] (c) If $\mathbf{c}$ is such that $[p, q]^T$ is optimal and the set of optimal solutions of $(\mathbf{P})$ is unbounded then in the optimal table of $(\mathbf{P})$ give all j 's such that $c_j - z_j = 0$. **If possible** give all the variables in the optimal solution of the Dual of $(\mathbf{P})$ which are equal to 0. Justify.

Soln: (a) The nonbasic variables are x_1, s_2 since these variables take the value 0 at $[p, q]^T$ and the basic variables are x_2, s_1, s_3 since each of these variables take positive value at $[p, q]^T$.

Sol: (b) There are no **degenerate BFS**.

A BFS is said to be degenerate if at least one of the basic variables in the BFS are equal to 0. Since there are three constraints other than the non negativity constraints and three variables s_1, s_2, s_3 are added, so in any BFS there are three basic variables and two nonbasic variables which take the value 0.

The feasible region has four extreme points each of which corresponds to a BFS and these are the only BFS.

(i) The extreme point $[p, q]^T$ which is the point of intersection of the LI defining hyperplanes $s_2 = 0$ and $x_1 = 0$ is nondegenerate since each of the basic variables x_2, s_1, s_3 take positive values (since the defining hyperplanes $x_2 = 0, s_1 = 0, s_3 = 0$ does not pass through this point).

(ii) The extreme point which is at the point of intersection of the LI defining hyperplanes $s_1 = 0$ and $x_1 = 0$ is nondegenerate since each of the basic variables x_2, s_2, s_3 take positive values (since the defining hyperplanes $x_2 = 0, s_2 = 0, s_3 = 0$ does not pass through this point).

(iii) The extreme point which is at the point of intersection of the LI defining hyperplanes $s_1 = 0$ and $s_3 = 0$ is nondegenerate since each of the basic variables x_1, x_2, s_2 take positive values (since the defining hyperplanes $x_1 = 0, x_2 = 0, s_2 = 0$ does not pass through this point).

(iv) The extreme point which is at the point of intersection of the LI defining hyperplanes $x_2 = 0$ and $s_3 = 0$ is nondegenerate and since each of the basic variables are x_1, s_1, s_2 take positive values (since the defining hyperplanes $x_1 = 0, s_1 = 0, s_2 = 0$ does not pass through this point).

Sol: (c) $c_j - z_j$ values corresponding to the basic variables x_2, s_1, s_3 are equal to 0. [.5] mark
 Since (P) has multiple optimal solutions along the ray starting from $[p, q]^T$ so $c_{x_1} - z_{x_1} = 0$ (explain why). [1.5] mark

Since both s_1, s_3 are strictly positive so by the complementary slackness theorem the corresponding dual variables $y_1 = 0, y_3 = 0$. [1] mark

However if $\mathbf{c} \neq \mathbf{0}$ then $y_2 > 0$, since $y_2 = 0$ would imply that all the feasible solutions in the region bounded by $[p, q]^T$ the adjacent extreme point and the ray starting from $[p, q]^T$ will all be optimal solutions which will imply $\mathbf{c} = \mathbf{0}$. [1] mark